

APPROVED METHOD:
**PHOTOMETRIC TESTING OF SKYLIGHTS AND
TUBULAR DAYLIGHTING DEVICES UNDER
HEMISPHERICAL SKY CONDITIONS**
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-81-20(R2023)

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Publication of this Committee
report has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared by:
IES Testing Procedures Committee**



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Approved by the IES Standards Committee, December 19, 2022, as a Transaction of the Illuminating Engineering Society.

Approved April 28, 2023, as an American National Standard.

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Published by the Illuminating Engineering Society, 120 Wall Street, New York, New York 10005.

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Printed in the United States of America.

ISBN# 978-0-87995-239-6

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1.0 Introduction and Scope

1.1 Introduction

Unlike electric luminaires, which require only one photometric test, skylights require a separate test to characterize their performance at different sun angles as well as a separate measurement to determine the sky conditions. Skylights are very often large; some exceed the size capabilities of typical intensity distribution goniophotometers. Therefore, a method was developed to quickly and accurately measure daylight data and place the results in a standardized photometric file format such as ANSI/IES LM-63-19.¹ The collection of results can then be used by lighting design software programs to simulate how skylights under the desired test conditions will fill a space with daylight. This equipment tested also proved to be accurate with regard to the effects of test distance.⁶

This is only one method of achieving luminous intensity distribution curves and efficiency data for skylights and tubular daylighting devices with diffusing properties. In addition, this method mirrors the proven method used to measure electric lights but includes many adjustments because of the very intense, movable light source, the sun. Since 2002, this method has proved to produce consistent repeatable results for measuring skylights and tubular daylighting devices under actual sky conditions.

The procedures contained in this document are a benefit to the lighting industry in characterizing the performance of daylighting products and are backed by significant experience using this measurement method. However, daylight photometry is in infancy stages of research, and further development of these methods will continue. In addition, further research is being conducted to compare measurements to simulation methods. Simulations of daylight system performance use predicted sky models from either the International Commission on Illumination (CIE) or IES and allow for the rapid comparisons of multiple design options or sky conditions. Further research will also be conducted to methods of achieving accurate luminous intensity distribution curves and efficiency data.

1.2 Scope

This Lighting Measurement (LM) document provides the IES recommended uniform method for determining and reporting the photometric characteristics of skylights and tubular daylighting devices that incorporate a means to diffuse the natural hemispherical daylight as the daylight passes through the daylighting system. It describes the procedures followed and the precautions observed in obtaining uniform and reproducible measurements of tubular daylighting devices and skylights with glass or plastic glazing. This document identifies the components and the structure type needed to adequately measure daylighting devices. The procedures, calibration of the equipment, and determination of sun angles and sky conditions are also discussed. This method is not recommended for daylight devices with clear glazing (see **Annex B**).

Additional information on subjects covered in this document may be found in:

- Illuminating Engineering Society. ANSI/IES LS-4-20²
- Illuminating Engineering Society. ANSI/IES LM-63-19¹

2.0 Test Distance

The goniophotometer and the building enclosure should be set up to measure the largest desirable device. The distance between the daylight luminaire and the photodetector should be sufficiently large that the inverse-square law (ISL) can be applied at the distance needed for the largest daylight luminaire to be tested by the goniophotometer. Smaller daylight luminaires can also be tested. It is required that the distance from the daylight luminaire to the photo detector be at least five times the largest luminous dimension of the daylighting luminaire.

For example, when measuring the luminous intensity of a 610 mm x 610 mm (24 in. x 24 in.) skylight, the longest distance would be the 864-mm (35-in.) diagonal. Therefore, the minimum test distance needs to be 5 times 864 mm (5 x 35 in.), which is 4.45 m (14.6 ft). For a tubular daylighting device with a circular daylight opening of 533 mm (21 in.) the minimum test distance would be 2.67 m (8.75 ft). (See **Figure 2-1**.)

3.0 Goniophotometer Basic Construction

When performing photometry of luminaires, a rotating-mirror goniophotometer is commonly used. This turns the light path to near-horizontal and substantially reduces space. However, a rotating-mirror system cannot be used with a skylight because of mechanical interference problems. In order to minimize the required volume for the goniophotometer, mirrors can be used to bend the light path, allowing for adequate test distance in a smaller space.

The examples contained in this document provide illustrations of some successful goniophotometer designs but are not the only applicable designs. These examples may not cover all of the eccentricities exhibited by a particular product or a specific design that might be needed to accommodate a particular type of device. New designs can be constructed as long as they meet or exceed the minimum guidelines specified within this document.

3.1 Small Aperture (Tubular) Goniophotometer

An example of a goniophotometer used to measure tubular daylighting devices 53 cm (21 in.) or smaller is illustrated in **Figure 3-1**. The frame shall be developed so

that all of the photodetectors are positioned to exceed the test distance described in **Section 2**. In addition, by positioning photodetectors in a 180-degree arc around the daylighting device, the goniophotometer will have to rotate less than 180 degrees. If the photodetectors are positioned from 0 to 85 degrees in one direction, the goniophotometer would have to rotate just less than 360 degrees. This would also increase the testing time. The goniophotometer should be developed to minimize the testing time. Additional information on reducing testing time can be found in **Section 3.8 Collecting Speed** and **Section 7.5 Testing Start Time in Relation to Sun Angle Times**.

3.2 Skylight Goniophotometer

Many standard and popular skylights are 1.2 x 1.2 m (4 x 4 ft). In order to meet the test distance, the equipment should be set up so that the distance between the skylight and the photodetectors is at least 5 times the diagonal. The diagonal of such a skylight is 1.7m (5.7 ft); thus, a test distance of 8.5m (27.9 ft) is needed. Therefore, the minimum room size that will allow data collection over a complete downward hemisphere is over 17 x 17 m (56 x 56 ft), with a ceiling height of over 8.5 m (27.9 ft). Such a size of chamber is not practical from the standpoint of cost and availability. Therefore, larger skylights can be measured in 4 equal quadrants and the results summed. For example, the nominal

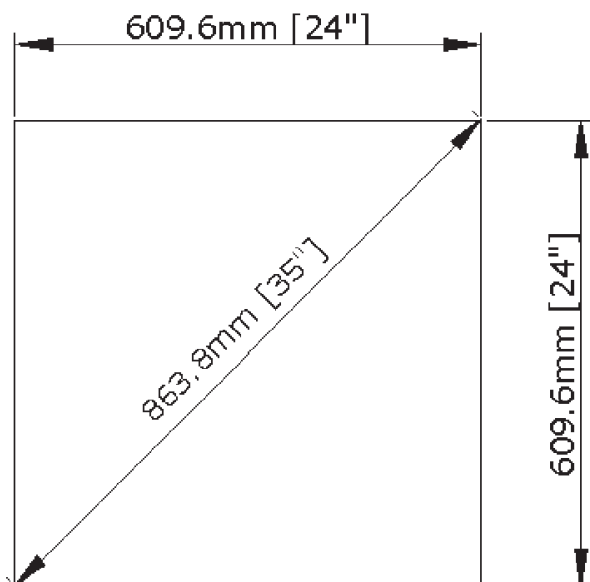
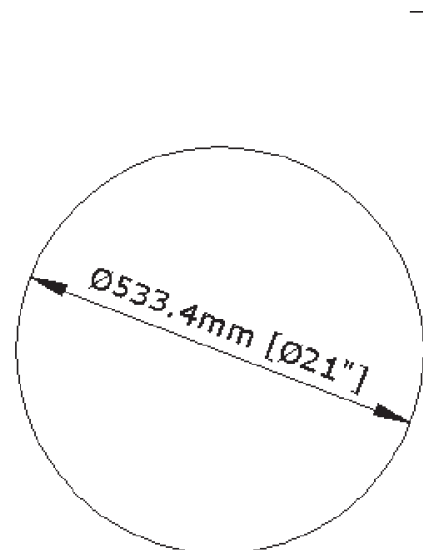


Figure 2-1. Finding the largest luminous opening.



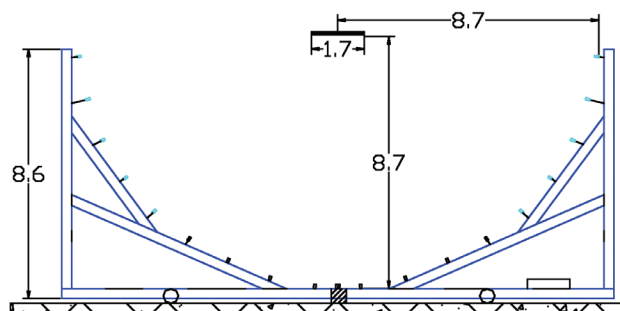


Figure 3-1. Example cross-section of a simple goniophotometer.

bottom opening of each skylight that is 1.2 x 1.2 m (4 x 4 ft) is 1.5 square meters (m²) (16 ft²). However, only one-quarter of this, or 0.61 x 0.61 m (2 x 2 ft), can be exposed to the goniophotometer at any one time. The remaining area is blocked by the ceiling. It is critical that the blocked light not be allowed to reflect from the roof back into the skylight well to any significant degree, because it may reflect again and be added to the light being measured. To avoid these reflections, a black card is laid over the three unexposed quadrants of the skylight's well bottom, on top of the roof, inside and at the bottom of the skylight well. This reduces inter-reflected light to a negligible level, as has been verified by measurements and calculations provided in **Annex A**. An example of a space-saving rotating photometer with the anti-reflective shields can be viewed in **Figure 3-2**. **Figure 3-3** shows an example of the aperture in the ceiling of the goniophotometer building.

3.3 Electrical/Electronic System

The reader is referred to other documents for guidance in specifying the photodetectors and suitable measuring circuits used in the goniophotometer.^{3,4}

Each photodetectors is a silicon cell, spectrally corrected to the CIE $V(\lambda)$ curve, with $f_1' < 2\%$. This photo detector should be used because it exhibits low noise, appropriate sensitivity, excellent stability and linearity, fast response, and low temperature sensitivity. The use of photo detectors and suitable measuring circuits are described in ANSI/IES LM-46-20.³

Ten or more photodetectors should be positioned, starting at the nadir point directly below the daylight luminaire and continuing in regular angular increments

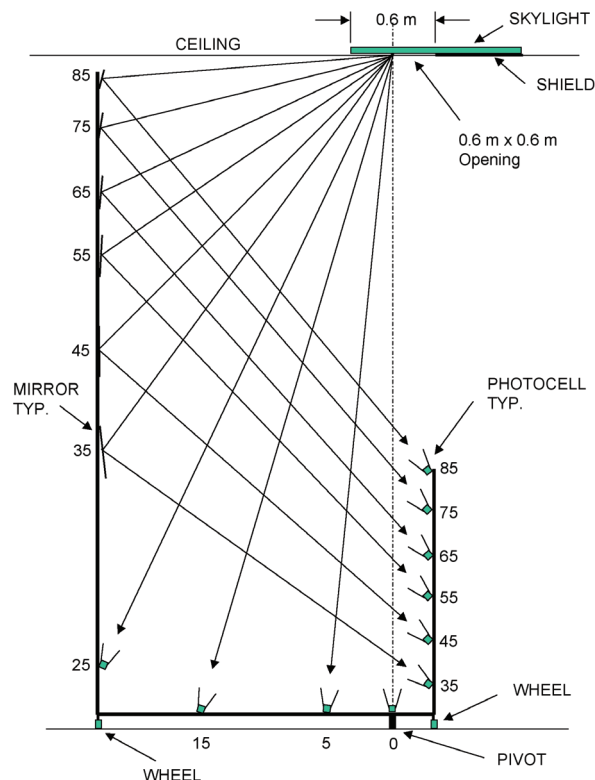


Figure 3-2. An example of a goniophotometer using mirrors to save space.

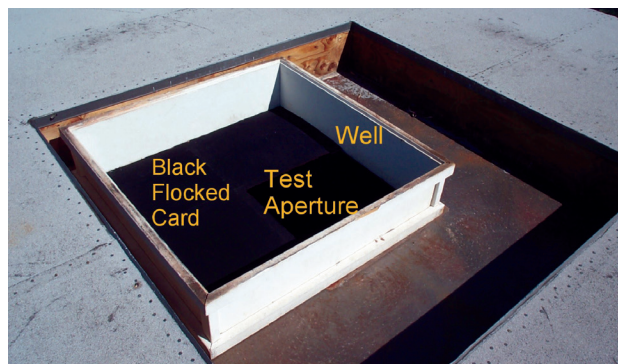
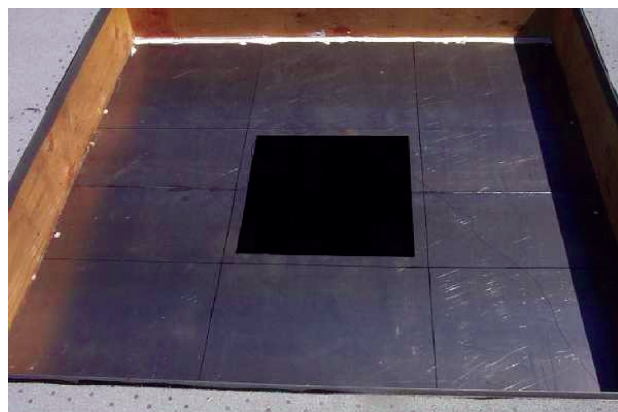


Figure 3-3. An example of the aperture in the ceiling and the means for blocking reflections from the roof.

toward the ceiling plane (see **Figure 3-2**). These photodetectors need to be supported by a rigid structure that can be rotated around the daylighting device to allow for the measurements of each angular plane or half-plane, depending of the size of the daylighting device. Black-painted 100-mm-square (4-inch-square) steel tubing provides an ideal choice of framing material for a photometer needed to measure a 1.2 x 1.2 m (4 x 4 ft) skylight when mirrors were used to reduce the size of the structure. Additional bracing and adequate frame size should be used to ensure that the structure is stabilized quickly after each movement to the next vertical plane.

3.4 Amplifier Box

An amplifier box should be positioned adjacent to each photodetector. The amplifier should be equipped with a switchable feed back resistor to provide high or low gain readings. The feedback resistors should allow measurements up to 350,000 candelas to be recorded. This sufficient to cover the range from direct sunlight to the low levels found with highly diffusing skylights. Thus, there are no limitations on the type of skylight than can be measured, apart from size. Each amplifier output runs to a main electronics interface box that is located on the horizontal rotating arm.

3.5 Electronic Interface Box

Because of the short-term variability of daylight, it is necessary to capture the results as quickly as possible by using sensors that are connected to a data logger with recording capability (ISNA 8-21). The amplified analog signal shall be converted to digital and sent to computer interface equipment. The entire system should be multi-channel such that the computer monitors all photocells simultaneously.

3.6 Computer Interface Board

Output from the electronics box runs to a computer equipped with a multiple-channel interface. This allows the software to read the signal generated by each amplifier.

The computer interface board is equipped with output channels. The software is able to send pulses to a stepper motor, thus causing rotation of the goniophotometer about its vertical axis to any desired horizontal angle.

If any photodetector, when operating in a high-gain condition, indicates an over-range condition, this is identified by the software. The computer then sends an output pulse to the amplifier boxes adjacent to the photodetectors, switching a relay that selects a different feedback resistor, thereby setting the gain stage to low.

The entire process may be automated, such that a complete test at any array of horizontal angles can be collected under computer control using high and/or low gain stages. The software automatically rejects out-of-range readings taken using the high-gain stage, and for those angles, substitutes the low-gain stage readings.

3.7 Vertical Angular Planes (Step Motor)

When measuring daylight devices, the speed of the test is very important. The number of measured planes shall be enough to accurately capture the luminous intensity while providing a quick testing time. At least sixteen planes of measurements taken every 22.5 degrees are recommend to achieve this luminous intensity curve if the goniophotometer can only measure half-planes (see **Figure 3-2**). When measuring smaller daylighting devices and using a goniophotometer that can measure horizontal angles in both directions, as in **Figure 3-1**, eight full planes of measurements every 22.5° are the required minimum number of vertical planes.

Vertical angular plane measurements should be every 22.5°, with a tolerance of $\pm 1^\circ$ for both location and repeatability. For example, a high-powered digital stepper motor could be used in conjunction with an appropriately sized solid rubber tire to position the goniophotometer at each vertical plane position. For the skylight photometer shown in **Figure 3-2**, the circumference of the wheel is 63 cm (2.0 ft); thus, it travels this distance in one revolution. The digital pulses per revolution are 50,800; therefore, each pulse corresponds to a linear movement of the wheel of 63/50,800, or 0.012 mm (0.0005 in.). The distance of the wheel from the central pivot point at the photometer center is 2.1 meters (7.0 ft). A single pulse relayed to the stepper motor therefore causes it to rotate (0.012/2,100) radians, or 0.0003°, which is therefore the horizontal angular resolution of the photometer.

For the procedure described here, the goniophotometer shall be aligned to North prior to taking measurements.

Aligning the goniophotometer back to the 0° or due North position can be achieved by placing a reflector on the wall of the building at the 0 degrees due North position and placing a laser on the leading edge of the goniophotometer so that when the goniophotometer is returning to the original position the laser will sense the reflector and cause the goniophotometer to stop at the 0° original home position.

3.8 Data Collection Speed

It is important for data collection to occur at high speed, such that the results shall not be significantly affected by movement of the sun during the test. Therefore, a multiple-cell goniophotometer should be used to capture the whole vertical plane reading nearly instantaneously.

One set of vertical-plane measurements are needed for tubular daylight devices and skylights that are smaller than the goniophotometer building's daylight opening. For example, a goniophotometer building with an opening of 0.61 x 0.61 m (2 x 2 ft) at the ceiling level can measure a tunnel diameter up to 0.61 m (2 ft), or a skylight that is 0.61 x 0.61 m (2 x 2 ft), with one set of vertical plane measurements. An ideal set of vertical-plane measurements could be captured by measuring the light output at every 22.5° around a 360° circle. However, if a skylight that is 1.2 x 1.2 m (4 x 4 ft) is being measured, four sets of measurements will be needed before all the results can be added together to arrive at the absolute results. Each set of measurements shall be achieved within 2 minutes. The goniophotometer then rotates in the opposite direction, back to its 0° reference plane. During this time, the skylight is repositioned for the second quadrant. This procedure is repeated until all four quadrants have been measured (see **Figure 3-4**). Further detail about testing times in relation to solar angles can be found in **Section 7.5**.

Four sets of measurements will be needed as described above if measuring tubular daylight devices while changing the orientation to face North, East, South, and West. Under these circumstances, testing shall start about 6 to 8 minutes prior to the time of the desired sun angle.

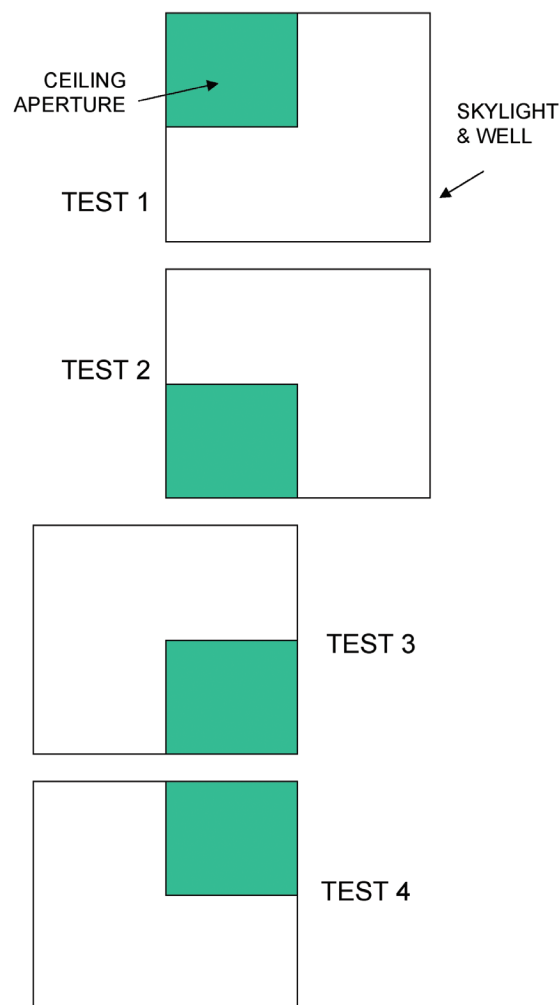


Figure 3-4. Testing in all four quadrants of the skylight.

3.9 Automated Goniophotometer

A photometric test system may be partially or fully automated through a computer. The linearity of the complete system should be maintained with frequent checks throughout its range.

4.0 Photometer Structure

A structure to house the goniophotometer shall be large enough for the goniophotometer, the data logging equipment, and the computer. In addition, all extraneous light (light other than that coming through daylighting device) needs to be blocked during the test. **Figures 4-1** and **4-2** show an example of a goniophotometer housing used to

measure 1.2 x 1.2 m (4 x 4 ft) skylights and smaller devices. The chamber floor area is 5.5 x 5.5 m (18 x 18 ft), and its actual height is 5.5 m (18 ft). The roof of the building should receive direct sunlight without blockage from other structures or trees.



Figure 4-1. The chamber floor area is 5.5 x 5.5 m (18 x 18 ft), and its actual height is 5.5 m (18 ft). The roof of the building should receive direct sunlight without blockage from other structures or trees.

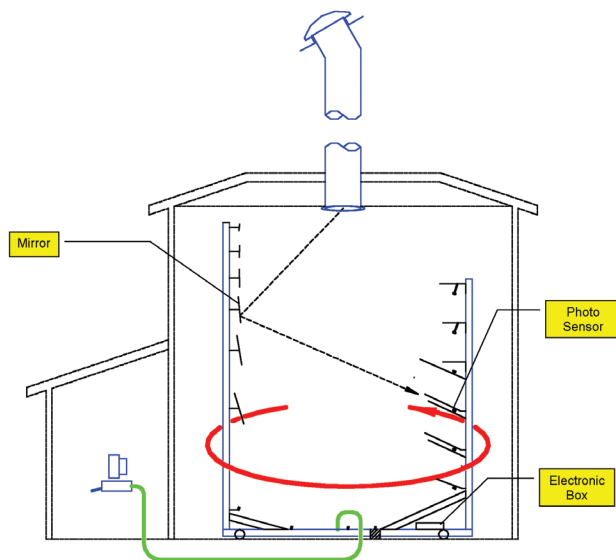


Figure 4-2. Section view of the goniophotometer building shown in Figure 4-1.

Rotating and tilting a building with the goniophotometer inside would allow the rapid simulation of solar elevations and azimuths. (One would not have to wait for the sun to move and would not be limited to time of

year to test.) However, the effects of a smaller sky dome still need to be investigated. Either goniophotometer type (see **Figures 3-1** and **3-2** in **Sections 3.1** and **3.2**, respectively) could be utilized with this structure design. The incident angle to the tube opening would determine the sun's angle, and the tube-floor-sensors geometry would remain constant.

5.0 Calibration

Prior to the daylighting device being positioned over the ceiling aperture, the goniophotometer should be calibrated.

5.1 Periodic Calibration and Accuracy

Because the accuracy of the data depends on the initial set up, particular attention shall be given to providing proper calibration and verifying accuracy. An absolute calibration is necessary and can be achieved by using the calibration fixture shown in **Figure 5-1**.



Figure 5-1. Example of a calibration fixture.

The aiming of the photodetectors and the positioning of the mirrors shall be verified and adjusted using a laser with a specially designed holder clamped into the photodetector holder. The laser should maintain contact with the center of the skylight diffuser or light shaft opening while the goniophotometer is rotated. All photodetectors and mirror positions shall be verified using the laser, to ensure that the entire set of angular coordinates of the measurement system are accurate.

To achieve calibration, a series of calibrating factors shall be developed, one factor for each photodetector, in the form of candelas per millivolt. These shall be stored in a data file and used to automatically convert the amplifier output voltage to absolute candelas.

A further factor to apply the gain stage change between high and low settings is fixed at x100 for each photodetector. This factor is dependent upon the feedback resistors used, which are precision rated to $\pm 1\%$.

The multiple-cell goniophotometer shall be calibrated with a special form of a standard lamp (see **Figure 5-1**). This special calibration fixture should provide a wide spread of light that is reasonably uniform with respect to changing vertical angles. This source is calibrated over a range of vertical angles with the lamp operating at a specific current, which is traceable to the U.S. National Institute of Standards and Technology (NIST). Intensity values were collected at each vertical angle of interest in the vertical plane perpendicular to the lamp's axis.

Ideally, a calibration source should be of about the same intensity as the light to be tested. This is difficult when the source to be tested is the sun. Therefore, care shall be taken to ensure that the calibrations between the scales on the system are very accurate. If one scale is 10x, it should be 10.00, not 10.01. This can be achieved using precision resistors in the amplifier circuit. Otherwise, if the calibration and testing are done, for example, 3 scales apart, the error can be multiplied and become significant.

An easy check the scale circuits of the goniophotometer is to run a photometric test using a properly calibrated illuminance meter placed next to a photocell. It should be verified that it gives the same reading as the photometric test report shows. In this way, the user knows that any scaling errors in the calibration are not a problem. Typically, there should be a $\pm 2\%$ difference, or less.

Since the photometer is measuring light coming through a skylight, the actual levels measured may only be 100, 200, 500 lux (approximately 10, 20, 50 footcandles), not

the 100,000 lux (approximately 10,000 fc) that direct sunlight produces. The theory which states that the calibration source needs to be equivalent to the source being measured is not nearly as great a problem as might initially be apparent.

The calibration source is placed at the center of the daylight opening at ceiling level. The surrounding surfaces are black. With the absolute intensity at each angle from the mirror goniophotometer data known, the calibration factor in candelas per millivolt was determined for each photodetector. It is important to note that effects due to differing test distances and possible differences in the reflectance of the various mirrors will cancel when these factors are applied to test readings.

5.2 Data Reduction

Data shall be developed into standard IES Files for each daylight device and each 10-degree elevation of the sun. Test reports shall be produced in a format similar to that used for indoor luminaires and shall include the information noted in **Section 7.6 Reporting**.

Modifications to existing software were required to handle a unique characteristic of skylights: they possess no axis of symmetry. Data presentation, therefore, shall be developed for all measured planes from 0 degrees (designated as North) to 337.5°, in steps of 22.5°.

As part of the data reduction, all data may be prorated to 1,000 lumens incident upon the upper plane of the skylight to form a constant reference basis (as with electric lights), or an absolute value may be outputted to the file.

5.3 Stray Light

Precautions shall be taken to eliminate stray light from the test environment by the use of adequate shielding and baffling. For a moving-photodetector goniophotometer, the light incident on the photodetector shall only be that which is directly transmitted from the daylighting device or mirror corresponding to a given photodetector. The presence of stray light shall be detected by blocking off the direct light from the source.

Black shields shall be installed between the photodetectors of the goniophotometer with mirrors to prevent stray light from being picked up by a sensor other than the intended photodetector.

To minimize the effects of stray light, the equipment and the walls, ceilings, and floor of the photometric test room shall be painted flat black or covered with matte cloth such as black velvet to eliminate reflected light.

Any remaining stray light shall be measured by running a complete test with the direct light from the luminaires completely shielded from the photodetector. This light is then subtracted from the data, taking into account the variations of stray light for each vertical angle in each plane or half-plane measurement.

6.0 Sky Conditions

The sky conditions shall be verified visually and recorded, and the IES standards shall be used to calculate the sky conditions. Skies are divided into three categories: clear, partly cloudy, and overcast. Calculation of daylight availability at a site begins with a determination of the solar position, which is a function of the latitude and longitude of the site, the day of the year, and the local time.

Either the sky-ratio method or the sky-cover method is used to classify a sky. The sky ratio is determined by dividing the horizontal sky irradiance by the global horizontal irradiance.

This shall be accomplished by positioning a calibrated, cosine-corrected illuminance meter near the top of the skylight in an unobstructed location that does not interfere with the light penetrating the skylight. If the skylight is being measured as it would be installed, the sky condition measurements (ratio) shall be taken while the meter is positioned parallel to the ground and facing upward. However, if a tilting goniophotometer is used, the illuminance meter should be tilted to coincide with the tilt angle of the goniophotometer.

Two readings are taken to check climatic conditions: one for total horizontal illuminance, and a second where the direct sunlight is screened from the photocell using a shield that only blocks the direct light from making contact with the sensor. A ratio is calculated for the total diffuse illuminance. This ratio is used to determine the sky conditions:

Clear: sky ratio ≤ 0.3

Partly cloudy: $0.3 < \text{sky ratio} < 0.8$

Overcast: $0.8 \leq \text{sky ratio}$

Since the sky ratio approaches 1.0 when the solar altitude approaches zero (regardless of the sky condition), this method is not accurate for low solar altitudes.

Screening the direct sunlight from the photocell was accomplished by holding a 76-mm (3-inch) diameter shield 460 to 610 mm (18 to 24 in.) from the photometer sensing element. The shield should be attached to a rigid, thin handle that allows for consistent measurements and does not interfere with the direct or indirect sunlight. The sky condition monitoring station shall be positioned in a location near top of the skylight being measured but shall not interfere with the direct or indirect light making contact with the skylight.

The ratio of horizontal diffuse to total horizontal illuminance shall be listed in the printed reports and in the IES files as "sky ratio."

Total horizontal illuminance readings shall be made at the same time that each plane of photometric data is collected. These readings shall be used to normalize the photometric data and to identify changes in the daylight conditions during the time of testing.

7.0 Testing Procedures

7.1 General Test Conditions

Skylights and tubular daylighting devices shall be clean and mounted according to the manufacturer's installation instructions prior to testing. When cleaning a daylighting device, it is desirable to return the daylighting device to as near the "like new" conditions

as possible, avoiding the use of corrosive or abrasive materials, which might alter the finish of the lamp, reflector, refractor, glass, or plastic.

Luminaire luminance characteristics are reported to facilitate evaluation of certain quality aspects of design. Maximum luminance is measured directly from the daylight luminaire, while average luminance is a quantity calculated from photometric data. The angle of the sun in relation to the daylighting luminaire will vary over time, and this will produce a wide range of results; daylight luminaires are not considered symmetrical. A complete set of vertical-plane angles are required to measure and accurately report the light distribution for daylight luminaires. In addition, the results from one sun angle are not valid for other sun angles.

7.2 Daylight Device Positioning

ANSI/IES LM-72-20 provides guidance regarding the directional positioning of photometric data.⁵

The daylighting device shall be properly positioned on the goniophotometer building to obtain correct luminous intensity distribution measurements at specific vertical angles and angular planes. It is critical that the daylighting device be centered over the nadir point.

Extreme care shall be taken that the operator or other personnel involved in the test proceedings do not cast shadows on or block diffuse or direct light from making contact with the product being measured. Access to the light measuring equipment room shall be limited or avoided during the time of testing to maintain stable dark readings. Mirrors shall be maintained and cleaned in order not to diminish the readings.

7.3 Sun Angles

Unlike electric luminaires, the photometric distribution of skylights change rapidly throughout the day, unless the sky is overcast to the extent that the sun component is almost totally obscured. Thus, the IES requirement for stabilization of the light output does not apply. Rather, the need is to perform testing as quickly as possible at a particular chosen angle of the sun. This needs to be conducted over a period where the sun movement can be considered negligible. Inevitably, once the test

is completed, the facility has to be ready to perform another test as the sun approaches the next selected angle.

It is desirable to conduct testing for every 10-degree increment of the sun's elevation over an entire day. **Annex C** provides the method used to determine the exact times at which the desired solar angle is reached. Testing should be conducted in a location with an abundance of clear sunny days and which experiences solar elevations of at least 75 degrees. The testing latitude needs to be considered in the calculations.

In order to determine whether 10-degree steps in elevation are satisfactory for interpolation for different solar altitudes, test data measured at solar altitudes of 20° and 40° was averaged. The resulting averaged data matched well with the results measured at a solar altitude of 30°. The resulting skylight efficiency of the averaged data is only 2% different from the actual test data. The maximum intensities of the averaged data and the actual data are in identical directions, with magnitudes that differ by less than 1%. Clearly, if averaging 20° and 40° data gives such a good 30° approximation, it can be expected that averaging 20° and 30° data will result in a reasonable 25° approximation. Collecting data for 10° altitude steps provides robust information that can be used for interpolation to find data applicable to intermediate solar angles.

The data for this interpolation were taken from tests of a white acrylic domed skylight with a white diffuse light well. It should be noted that the results of averaging data from more-transparent skylights and more-specular light wells might not result in similar approximations. These comments have been made regarding solar azimuth⁶:

"As an economy measure, we have made the simplifying assumptions that skylight luminous intensity distributions under clear skies are primarily a function of solar altitude and azimuth. Implicit in this statement is that the relative solar azimuth to that of the skylight orientation is not important (e.g., if we rotate a skylight the distribution under the skylight does not change significantly). We know that this assumption is not true for some directional materials such

as structured polycarbonate but that it is a reasonable approximation for clear skylights and dome skylights with relatively isotropic diffusing glazings.

"Given the above assumptions, if one wanted to model a skylight under sunlight coming from a particular angle, one would need only pick the skylight photometrics tested under a solar elevation close to the modeled solar elevation. The solar azimuth in the model and photometric file can be aligned by rotating the photometrics azimuthally. This is accomplished by rotating the skylight in the lighting simulation software."

However, in order to accurately capture the effects of any optical geometry, photometric measurements should be measured at several azimuths for each corresponding sun angle. A good practice would be to position the daylighting device as it would be measured or installed and then rotate the skylight 30, 45, 60, and 90 degrees from the normal position to measure the effects of the optical elements.

The ambient temperature of the photometric daylight laboratory does not have to be maintained at a certain level for accurately measuring the daylight produced by a particular product. However, air quality for the comfort of the employees may have some benefits.

7.4 Selection of Angles for Photometric Measurements

7.4.1 Angles of Measurement. The vertical increments around the source or luminaire under test shall be small enough so that the recorded data describe the light distribution adequately. Within each vertical half-plane, intensity is measured at a minimum of these vertical angles: 0, 5, 15, 25, 35, 45, 55, 65, 75, and 85 degrees.²

7.4.2 Planes for Measurement. Because the angle of the sun in relation to the daylighting luminaire will vary over time, producing a wide range of results, daylight luminaires are not considered symmetrical. A complete set of horizontal angles is required in order to measure and accurately report the light distribution from daylight luminaires. In addition, the results from one sun angle are not valid for other sun angles.

The number of vertical-plane angles at which the readings are taken will depend on the rate of change in the distribution of the luminaire's luminous intensity. A complete set of vertical-plane angles is needed to capture the results for the total 360-degree luminous intensity distribution curve. To satisfy these requirements, a minimum set of 16 vertical-plane measurements needs to be recorded, at a minimum of every 22.5°, with the 0° and 360° measurements being the first and last set of measurements, respectively.

The test measurements for one sun angle may not be used for another sun angle.

Angular relationships shall be considered as follows:

- Vertically: Zero degrees at nadir; 90 degrees at horizontal
- Laterally: Measured counterclockwise

7.5 Testing Start Time in Relation to Sun Angle Times

In order to achieve an angular measurement at a given sun angle, measurements shall be started prior to the desired sun angle in half the time it takes to capture a complete set of measurements for a daylight device. For example, if a tubular daylighting device or a skylight smaller than the daylight opening in the photometer building is being measured, and the process time to complete one set of measurements is 2 minutes, then the test should be started 1 minute prior to the desired sun angle time. For a larger skylight requiring four sets of tests to capture all the quadrants, and assuming a process time of 2 minutes for each measurement (for a total 8 minutes total), the test should be started 4 minutes prior to the desired sun angle time. (Refer to **Figures 3-3** and **3-4** in **Sections 3.2** and **3.8**, respectively, to see how the quadrants of a larger skylight can be measured separately.)

The entire test for a 1.2 x 1.2 m (4 x 4 ft) skylight requires 12 minutes. In the summer, this equates approximately to a change in solar elevation of $\pm 3^\circ$. By starting the test 6 minutes prior to the desired solar angle being reached, the sun's position for all data collection is accurate to $\pm 1.5^\circ$. It is important to note that any errors introduced by this small angular change would tend to be self-canceling when the tests for the four quadrants are added together, because of the plus and minus angles.

Tubular daylighting devices shall be measured within 2 to 3 minutes or a $\pm 1^\circ$ change in solar angle, and skylights shall be measured within 8 to 10 minutes or a $\pm 3^\circ$ change in solar angle. In all cases, the shorter the measuring process time, the better.

7.6 Reporting

For information on a practical format and terminology for presenting the information obtained from a typical photometric test, the reader is referred to ANSI/IES LS-4-

20, *Lighting Science: Measurement of Light – The Science of Photometry* (see **Section 1.2 Scope**).

The subject matter of the photometric test report may be logically divided into three categories: General information, Photometric Data, and Application Engineering Data.

In addition, a photometric report and output file shall include the following if applicable for the daylighting device:

Skylight Description:

[_SKYLIHT or TDD] skylight or TDD detailed description
[Major _AXIS] ###.# in degrees (default is zero, 0 = major axis N-S)
measured clockwise looking down
[EXTERIOR _GLAZING _TYPE] glazing description
[_INTERIOR _DIFFUSER _TYPE] diffuser description
[_GLAZING _Vt] 0.## visible transmittance of glazing material at
normal incidence
[_GLAZING _HAZE] 0.## haze of glazing material
[_UNIT _TYPE] Feet or Meters
[_GLAZING _AREA] ##.# horizontal projection of glazing area
[_DAYLIGHT _DEVICE _WIDTH] frame width or tunnel width
[_DAYLIGHT _DEVICE _DEPTH] frame depth or tunnel depth
[_DAYLIGHT _DEVICE _LENGTH] Light tunnel length or tunnel length

Test site time conditions:

[_Test _Long] ###.# longitude of the test location
[_Test _Lat] ###.# latitude of the test location
[_TEST _ELEV] ##.# latitude of the test location
[_TEST _DATE] mm/dd/yyyy
[_TEST _TIME] standard local time of test
[_TEST _TZ] Time zone of test location relative to the Greenwich
Meridian (EST = 5, PST = 8)

Sun and sky conditions:

[_SOLAR _ELEVATION] ##.# degrees
[_SOLAR _AZIMUTH] ##.# degrees (+ West of South. - East of South)
[_SKY-Ratio] 0.## ratio of diffuse to total horizontal
[_SKY _CONDITION] clear, partly cloudy, or overcast

8.0 Testing Procedure

Step 1: Calibrate goniophotometer.

Step 2: Close off daylight opening and read dark room data. Subtract stray light from the measured results. These measured results shall be under 1% of the value measured during product testing.

Step 3: Install skylight or TDD.

Step 4: Determine sun angle times.

Step 5: Based on the number of tests needed for each sun angle and the amount of time needed for each test, determine the total number of minutes needed, and then divide by 2 to arrive at the number of minutes to start the testing prior to the sun reaching the desired angle.

Step 6: Measure sky readings; calculate and record the sky ratio.

Step 7: Start testing and recording the horizontal illuminance for each reading.

Step 8: If necessary, rotate or move the skylight or TDD, and continue testing.

Process reports:

Step 9: Open the PSO files and add the corresponding horizontal illuminance values.

Step 10: Process the PSO files to create IES files per the guidelines established in this LM and within the guidelines established in ANSI/IES LM-63-19.¹

Step 11: Print the report and save the ".ies" file.

9.0 Limitations of Skylight Photometry

IES has developed numerous documents in its Lighting Measurement (LM) series that cover the photometry of most forms of lighting devices. While these describe broad principles that can be applied to skylight photometry, there is no definitive recommended practice for such work. Moreover, there are aspects of the photometry of skylights that pose unique problems.

- Skylights form part of the building, and during testing, they are required to be illuminated by

daylight. The daylight incident upon the upper surface of a skylight consists of that from the moving sun plus a significant sky component. The incident light, therefore, is by no means directionally uniform; duplicating its form by the use of some type of solar simulator would be very difficult and expensive. Therefore, a test system was developed in which the skylights operated under their natural conditions, being part of a building structure and illuminated by the natural source.

- Skylights are very often large and exceed the size capabilities of a typical intensity distribution goniophotometer. As the goal of the program was to be able to collect accurate data for recording in standard IES photometric files, the test system would need the ability to collect intensity readings over an entire downward hemisphere. Further, the equipment would need to be accurate with regard to the effects of test distance.⁷ The system developed meets the IES rule of thumb of providing a distance from the skylight center to the photodetector of at least five times the largest dimension of the skylight.
- Unlike electric luminaires, the photometric distribution of skylights changes rapidly throughout the day, unless the sky is overcast to the extent that the sun component is almost totally obscured. Thus, the requirement for stabilization of the light output from the luminaire does not apply. Rather, the need is to perform the testing as quickly as possible for a particular chosen angle of the sun. This is required to be conducted over a period where the sun movement can be considered negligible. Inevitably, once the test is complete, the facility will have to be ready to perform another test as the sun approaches each next selected angle.

These special requirements led to the development of a unique photometric test system

Annex A – Accuracy of the Quadrant Testing Method

The IES rule of thumb is that photometric errors are negligible as long as the test distance is more than five times the maximum dimension of the luminaire. By breaking a 1.2 x 1.2 m (4 x 4 ft) skylight into 4 equal individual areas, the largest dimension of any test item will be less than one-fifth of the test distance of the goniophotometer described. Therefore, this does not introduce any error related to source size greater than that which occurs in conventional photometry. It is, in fact, equivalent to doubling the test distance.

It has been queried what error this technique may introduce if different portions of the skylight produce different levels of intensity in given directions. Case 1 in **Figure A-1** shows where a 1.2 x 1.2 m (4 x 4 ft) skylight is tested at 8.5 m (28 ft). This illustrates measurement at angle A . The skylight subtends an angular range B , and the intercepted vertical angles range from $A1$ to $A2$. The left and right portions of the skylight, marked *LHS* and *RHS*, may be considered equal. For the sake of this example, it may be assumed that *LHS* projects an intensity of 10 candelas (cd) toward the photocell, and

RHS projects 100 cd. It may also be assumed for this example that the intensity from both sides decreases with increasing vertical angle.

The angular range projected for *LHS* to the photocell ranges from A to $A1$. This light is assigned to an angle A by the software. All light from *LHS* is at angles lower than A , and therefore this light will be recorded and assigned to angle A at too high in intensity.

Similar logic applies to *RHS*, except that its light assigned to angle A will be too low in intensity.

Because light emission from *LHS* is much less than from *RHS*, the overall error will result in too low an intensity being assigned to angle A .

Cases 2 and 3 in **Figure A-1** illustrate where the photometry is performed separately on the two sections at a test distance that is half that used for Case 1.

In Case 2, the light assigned to A will actually be emitted over a range from $A1$ to $A2$. Provided that the gradient of intensity change with angle is reasonably close to constant, the increased emission in the range A to $A1$ will be compensated by the decreased emission over the range A to $A2$. This will greatly reduce any errors.

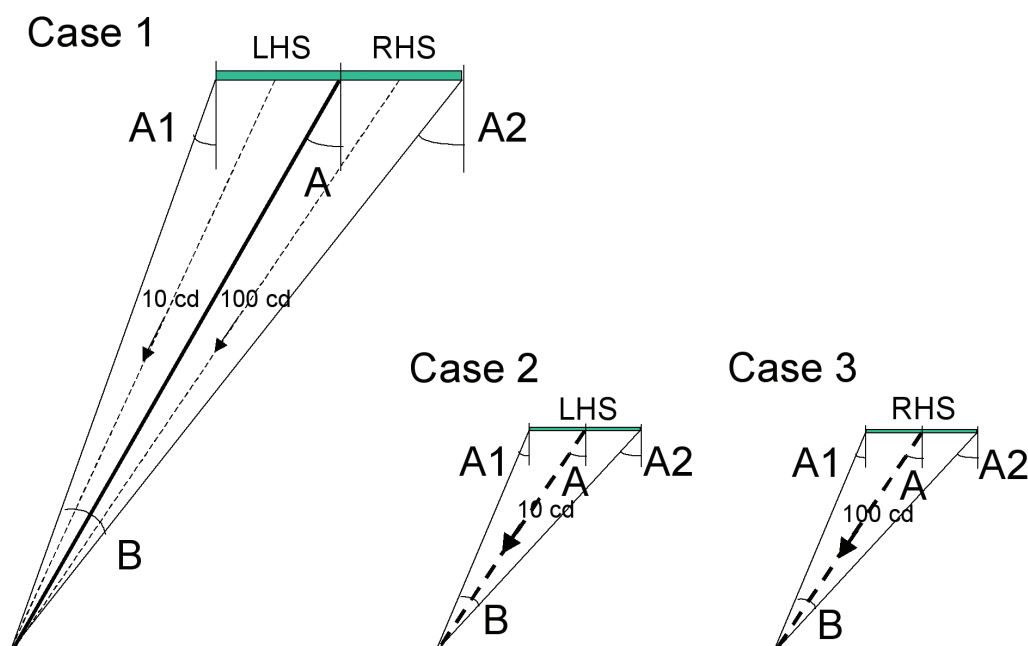


Figure A-1. Example illustrations of skylight angles.

In Case 3, the same error cancellation occurs.

When the readings from Cases 2 and 3 are summed, errors will be low because of the self-canceling tendency in the two individual cases. No such self-cancellation occurs in Case 1. The summation method, Case 2 + Case 3, is therefore inherently more accurate than Case 1. Case 1 meets the requirements of the 5-times rule; therefore, the Case 2 + Case 3 method exceeds the 5-times rule requirements.

It is therefore logical to assume that errors introduced by spatial non-homogeneity of a 1.2 x 1.2 m (4 x 4 ft) skylight will be reduced by recording values in four quadrants, rather than the errors being increased.

Where intensity gradients change sharply with angle, the self-canceling in Cases 2 and 3 will be imperfect, but there will always be some self-cancellation of errors. In such situations, there is a possibility that errors will increase with the Case 2 + Case 3 method as compared with the Case 1 method, as the angular range of measurement for Cases 2 and 3 is from A1 to A2, while that for Case 1 is half that amount. Sharp changes in intensity gradient, however, also are likely to introduce further errors in Case 1.

Errors in the application of the skylight tests will exceed those from conventional luminaire tests to the degree that such non-homogeneities are greater in the skylights. The extent of this effect is not known.

Annex B – The Testing of Transparent Skylights

The testing of *transparent* skylights is not recommended because the inverse-square law is invalid for these products. A thought experiment may be conducted to demonstrate why this is so: Imagine the sun shining directly through a tinted skylight at 4.5 m (15 ft) above a 0° photocell (looking straight up). Assume that the illuminance on the photocell is 64,500 lux (about 6,000 fc). Software will use the inverse-square law to calculate that this is due to a source intensity of 1.5×10^6 cd.

If someone were to then take the IES file generated from this situation and set up an application where the skylight was only 3 m (10 ft) above the floor, this would result in a calculated illuminance of 161,400 lux (about 15,000 fc). This is brighter than broad daylight.

The calculation is wrong because, for a *transparent* skylight, the light source is the sun, not the skylight. The illuminance measured directly below a skylight 4.5 m (15 ft) above the floor will be the same as the illumination on the floor. This is true because the sun is about 150 million kilometers (about 93 million miles) away, and a change of 4.5 m is not enough to make a detectable difference in illuminance. The transparent skylight is effectively a form of filter over the photocell.

Only *translucent* skylights transmitting absolutely no un-scattered light from the sun will fully obey the inverse-square law and are qualified to be tested using this method.

Annex C – Accuracy Analysis of Reflections from Black Card

Errors likely to be generated by the use of black flocked card over the three quadrants not being measured (refer to **Section 3.2 Skylight Goniophotometer**) were examined in detail before use of the technique. Three assumptions were made in the analysis.

- The material is a black flocked surface, reflectance = 2.2%, as measured by a reflectometer in accordance with *IES LM-44-1990, Approved Method for Total and Diffuse Reflectometry*.
- The pattern of reflected light is Lambertian.
- The average depth of the skylight well is 0.9 m (3.0 ft).

It is assumed that 100 lumens reach the 1.2 x 1.2 m (4 x 4 ft) bottom of the well. Lumens intercepted by the shield, on average, = 75 lm, as 1.1 m² (12 ft²) out of 1.5 m² (16 ft²) are shielded.

By computing the intensity distribution from a given point on the shield, assuming a Lambertian diffuser, and the solid angle subtended by the skylight itself (top

aperture), and the solid angle subtended by the skylight well surfaces, the proportion of reflected light from the point striking the well surfaces could be calculated. Computations indicated that for a typical point on the quadrant masking shield, 75% of the reflected light would strike the skylight well, and 25% would strike the skylight itself from underneath.

Thus, for the 75 lumens incident on the shield top surface, the lumens reflected and then intercepted by the well surfaces = $75 \times 0.022 \times 0.75 = 1.24$ lumens.

For a ray striking the well surface, which has a diffuse reflectance of 85%, 50% would reflect upward, 50% downward.

Amount of light reflected downward = $1.24 \text{ lm} \times 0.85 \times 0.5 = 0.53$ lumens.

Of this, one-quarter is likely to be transmitted through the aperture (0.4 m^2 [4 ft²] open out of the 1.5 m^2 [16 ft²] total).

$$\begin{aligned} \text{Amount of light reflected through opening} = \\ 0.25 \times 0.53 \text{ lm} = 0.13 \text{ lm} \end{aligned} \quad (\text{A-1})$$

Of the 100 lumens that reach the bottom of the well, the lumens reflected from the shield and intercepted by the skylight itself = $75 \text{ lm} \times 0.022 \times 0.25 = 0.41$ lumens.

The amount of light reflected upward from the skylight well surfaces (after reflection from the shield) = 0.53 lumens. (Identical to light reflected downward), assuming the worst case, that all of this reaches the skylight.

Therefore, the total light striking the bottom of the skylight itself after reflection from the shield top = $0.41 \text{ lm} + 0.53 \text{ lm} = 0.94$ lumens.

The reflectance of a sample skylight has been measured to be 7%. However, for the purpose of this calculation a more conservative assumption is made: a more diffuse skylight with a higher reflectance of 20%, as a worst case. The amount of light reflected by the skylight after reflecting from the shield top = $0.94 \text{ lm} \times 0.20 = 0.19$ lumens.

The proportion of this light reaching the $1.2 \times 1.2 \text{ m}$ (4 x 4 ft) bottom of the skylight well = 0.25 (using the same 75%/25% logic as earlier).

Of this, one-quarter will be transmitted through the $0.6 \times 0.6 \text{ m}$ (2 x 2 ft) unblocked aperture.

Therefore, light reflected upward from the shield top and then back downward from the skylight itself is:

$$0.19 \text{ lm} \times 0.25 \times 0.25 = 0.01 \text{ lumens} \quad (\text{A-2})$$

From **Equations A-1** and **A-2**, the total light transmitted through the $0.6 \times 0.6 \text{ m}$ (2 x 2 ft) aperture after reflecting from the shield top = $0.13 \text{ lm} + 0.01 \text{ lm} = 0.14$ lumens.

When data for the four quadrants are summed, the total error is 0.56 lumens.

The approximate error introduced by reflection from the shield is therefore 0.56%, as the above calculations are for 100 lumens reaching the bottom of the $1.2 \times 1.2 \text{ m}$ (4 x 4 ft) well.

The error introduced from light reflecting of the masking shield is considerably less than the expected overall measurement error, and it was felt that no adjustment factor was needed to correct for the presence of the shield.

Annex D – Calculation of Standard Time for Different Solar Elevations

Computing the standard time that the sun will be at certain specific elevations above the horizon is accomplished by first computing the *declination* and *right ascension* of the sun. Declination and right ascension are the equivalents of latitude and longitude on the celestial sphere. Next, the *local sidereal time* (LST) for the time of observation is computed. The LST is the right ascension of the Prime Meridian. The Prime Meridian is the great circle that runs through the zenith and the celestial poles. Once the celestial coordinates of the sun and the LST of the observation time are known, the hour angle of the sun can be computed.

The *hour angle* of any celestial object is the angle between the Prime Meridian and the great circle that runs through the object and the celestial poles. Hour angle and right ascension are normally expressed in terms of hours, minutes, and seconds. A full circle of 360° is converted to 24 hours; thus, 15° are equivalent to 1 hour. The elevation and azimuth of the sun can then be computed given the declination and hour angle of the sun, and the observer's latitude.

A BASIC language program was written incorporating two routines from outside sources and linked together with code written in-house. The two routines were for computing the declination and right ascension of the sun for a given calendar date and universal time, UT, and for computing LST from the calendar date, UT, and observer's longitude. Once the celestial coordinates of the sun, LST, and observer's latitude and longitude are known, the elevation and azimuth of the sun can be computed.

First, the hour angle of the sun needs to be computed. The relationship between right ascension *RA*, hour angle *HA*, and local sidereal time *LST*, is given by:

$$HA = LST - RA$$

Once the sun's hour angle is known, the solar elevation can be computed. The relationship between an object's declination δ , the observer's latitude ϕ , the object's zenith distance z , and the object's hour angle H , is given by⁸:

$$\cos z = \sin \phi \sin \delta + \cos \phi \cos \delta \cos H$$

The zenith distance z is related to the elevation e by:

$$z = 90^\circ - e$$

Therefore, the equation for elevation in terms of the hour angle is:

$$e = 90^\circ - \cos^{-1}(\sin \phi \sin \delta + \cos \phi \cos \delta \cos H)$$

Finally, the sun's azimuth can be calculated. The relationship between an object's declination δ , the observer's latitude ϕ , the object's zenith distance z , and the object's azimuth A is given by⁷:

$$\sin \delta = \sin \phi \cos z + \cos \phi \sin z \cos A$$

Solving the equation for azimuth gives:

$$A = \cos^{-1} [(\sin \delta - \sin \phi \cos z)/(\cos \phi \sin z)]$$

Computing the standard time at which the sun's elevation would be a multiple of 10° was accomplished by computing the solar elevation at one-minute intervals over the course of the entire day. The times for which the solar elevation is a multiple of 10° could then be determined to an accuracy of ±1 minute. The software was verified against computations made using the U.S. Naval Observatory website.

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Order# ANSI/IES LM-81-20(R2023)

ISBN# 978-0-87995-239-6

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